# Linear Maps, Change of Basis, Similarity

Math 5250 — Week 1

# Coordinates Relative to a Basis

Let $L : \mathbf{R}^n \to \mathbf{R}^m$ be linear, with bases $E = \{e_1, \ldots, e_n\}$ of $\mathbf{R}^n$ and $F = \{f_1, \ldots, f_m\}$ of $\mathbf{R}^m$.

- Writing $x = \sum x_i e_i$, there are unique scalars $y_1, \ldots, y_m$ with $L(x) = \sum y_i f_i$.
- In coordinates, $x = [x]_E$ and $L(x) = [L(x)]_F$.
- $L$ is determined by its values $L(e_i)$ on the basis $E$.

# The Matrix of a Linear Map

**Definition.** $[L]_E^F$ is the $m \times n$ matrix whose $i$-th column is $L(e_i)$ expressed in the basis $F$.

By construction,

$$[L(x)]_F = [L]_E^F \, [x]_E .$$

# Change of Basis

Let $E'$, $F'$ be other bases of $\mathbf{R}^n$, $\mathbf{R}^m$, and write $e_j = \sum_i a_{ij} e'_i$, giving a matrix $A = (a_{ij})$.

- $A[x]_E = [x]_{E'}$, so $A = [I]_E^{E'}$, the matrix of the identity map from basis $E$ to basis $E'$.
- $[I]_E^{E'} = \left([I]_{E'}^E\right)^{-1}$.
- The matrices of $L$ in the two pairs of bases are related by

$$[L]_E^F = [I]_{F'}^F \, [L]_{E'}^{F'} \, [I]_E^{E'}.$$

# Similarity of Matrices

For $L : \mathbf{R}^n \to \mathbf{R}^n$ and bases $E$, $E'$,

$$[L]_E^E = P[L]_{E'}^{E'} P^{-1}, \qquad P = [I]_{E'}^E.$$

**Definition.** Square matrices $A, B$ are **similar** if $A = PBP^{-1}$ for some invertible $P$.

- All matrices $[L]_E^E$ representing a single linear map $L$ (in different bases) are similar.
- **Proposition.** Similar matrices have the same rank and the same nullity. Rank and nullity are therefore well-defined invariants of $L$, independent of the basis.

# Injective, Surjective, and Isomorphisms

- $L : V \to W$ is **injective** if and only if its nullity is 0.
- $L : V \to W$ is **surjective** if and only if its rank equals $\dim W$.
- An **isomorphism** $L : V \to W$ is a linear map with a linear inverse; equivalently, $L$ is bijective, which forces $\dim V = \dim W$.

# Invertibility Criteria

**Proposition.** A linear map $L : \mathbf{R}^n \to \mathbf{R}^n$ is invertible if and only if its nullity is 0.

- An $n \times n$ matrix $A$ is invertible if and only if $\mathrm{rank}(A) = n$, equivalently $\mathrm{nullity}(A) = 0$.